Session 2

NestJS Framework

Objectives

By the end of this session, You will be able to:

1. Understand why NestJS exists
2. Learn the difference: Node.js vs Express vs NestJS
3. Master Decorators
4. Understand Separation of Concerns
5. Implement Middleware & Guards

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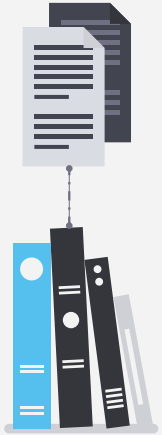

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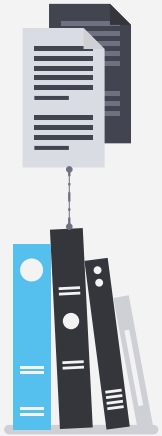

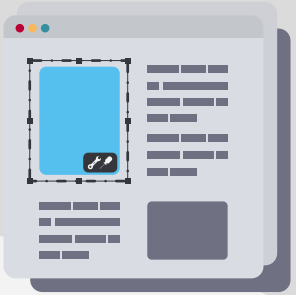

01 Recap

- What happened last Session?
- What did we talk about?
- What was the benefit you got from last session?

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1_ Problem with JS

2_ How the Problem is Solved

3_ What is TypeScript

4_ Basics of TypeScript

- ★ Types
- ★ Functions
- ★ Interfaces
- ★ Union Types
- ★ Intersection Types
- ★ Generic Types
- ★ Any
- ★ Type vs Interface vs Class vs Object

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5_ code Structure Different Philosophies

6_ How TS is Transpiled

7_ How to set it up

8_ What Files Matters to us

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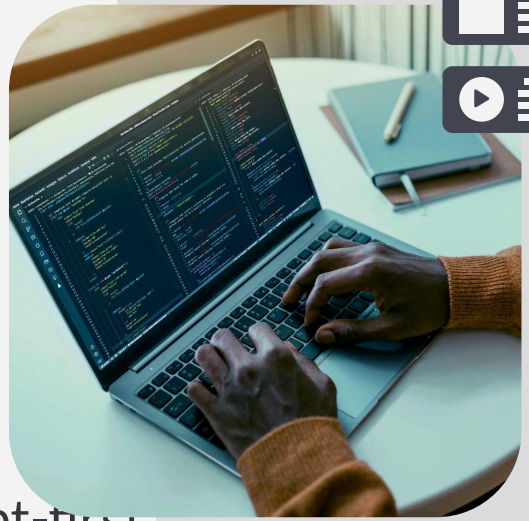
The Evolution & Why NestJS

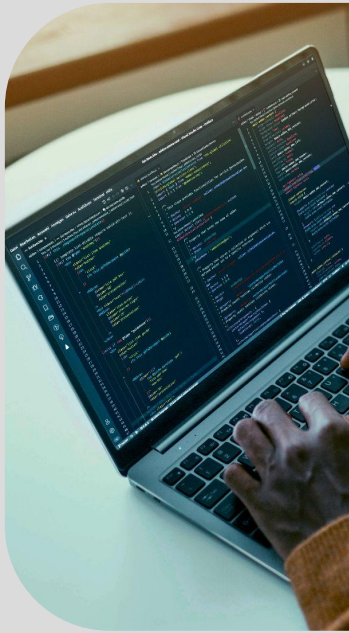
The Node.js Framework Timeline

- 2010: Express.js → Minimalist, unopinionated, flexible
- 2013: Koa.js → Async/await focus
- 2011: Hapi.js → Configuration-centric
- 2016: Fastify → Performance-first
- 2017: NestJS → Enterprise, opinionated, TypeScript-first



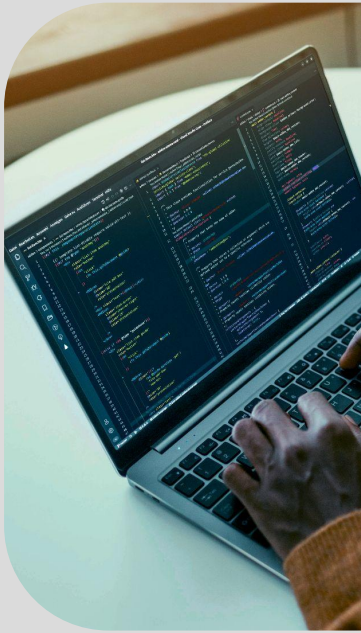
Express Problems at Scale

- As **projects** grow, Express applications often face several challenges:
- No enforced structure → Code becomes messy
- No standard patterns → Every developer does it differently
- Manual dependency management → Hard to test
- Mixing concerns → Business logic in route handlers



What is NestJS

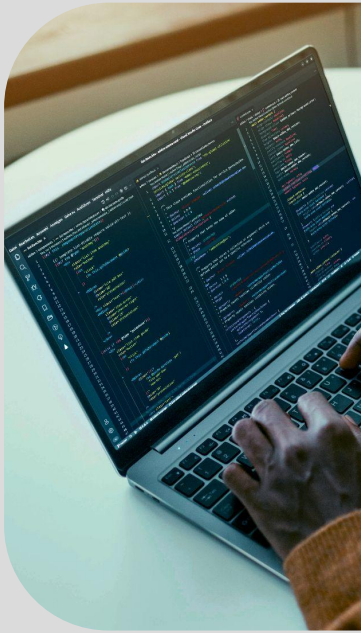
- a progressive opinionated Node.js framework for building efficient, scalable, and reliable server-side applications using TypeScript or JavaScript



Why NestJS Was Created

NestJS solves these problems by providing:

- Structure Chaos → Enforced modular architecture
- Scalability Issues → Built for large teams and projects
- Testing Overhead → Built-in testing utilities
- Dependency Hell → Automatic dependency injection
- Inconsistent Patterns → Standardized approach



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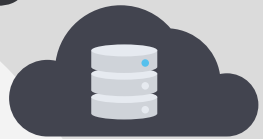
Node.js vs Express vs NestJS

Node.js vs Express vs NestJS



Understanding the Layers

- NestJS Application Code (Top Layer)
- NestJS Framework
- Express (or Fastify) Framework
- Node.js Runtime
- V8 JavaScript Engine (Bottom Layer)

Node.js vs Express vs NestJS

Node.js

- Runtime environment (like Java JVM)

Express

- Minimalist web framework

NestJs

- Opinionated framework built on Express/Fastify

Node.js vs Express vs NestJS

Aspect	Express	NestJS
Structure	None	Enforced modules
Dependencies	Manual imports	Dependency Injection
TypeScript	Optional	First-class
Testing	Manual setup	Built-in
Learning Curve	Easy	Steeper
Best For	Small apps	Enterprise apps

04

NestJS Philosophy

Core Principles

- ## TypeScript-First

  NestJS is built with and for TypeScript, catching errors at compile-time rather than runtime.

- ## Dependency Injection

  The framework manages dependencies for you, making code cleaner and easier to test.

- ## Modularity

  Code is organized into feature modules (e.g., Users, Products), keeping related code together.

- ## Platform Agnostic

  NestJS can run on top of Express (default) or Fastify, allowing you to switch underlying frameworks easily.

Architectural Inspirations

- ★ ***Angular*** → Module system, DI, decorators
- ★ ***Spring (Java)*** → Enterprise patterns, annotations
- ★ ***ASP.NET Core*** → Middleware pipeline, service registration

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Understanding Decorators

What Are Decorators?

- ***Definition:***
  Functions that attach metadata to classes, methods, or parameters. They are the "magic" that makes NestJS work.

- ***Think of them as:***
  Labels or tags that give special meaning to code.

Types of Decorators

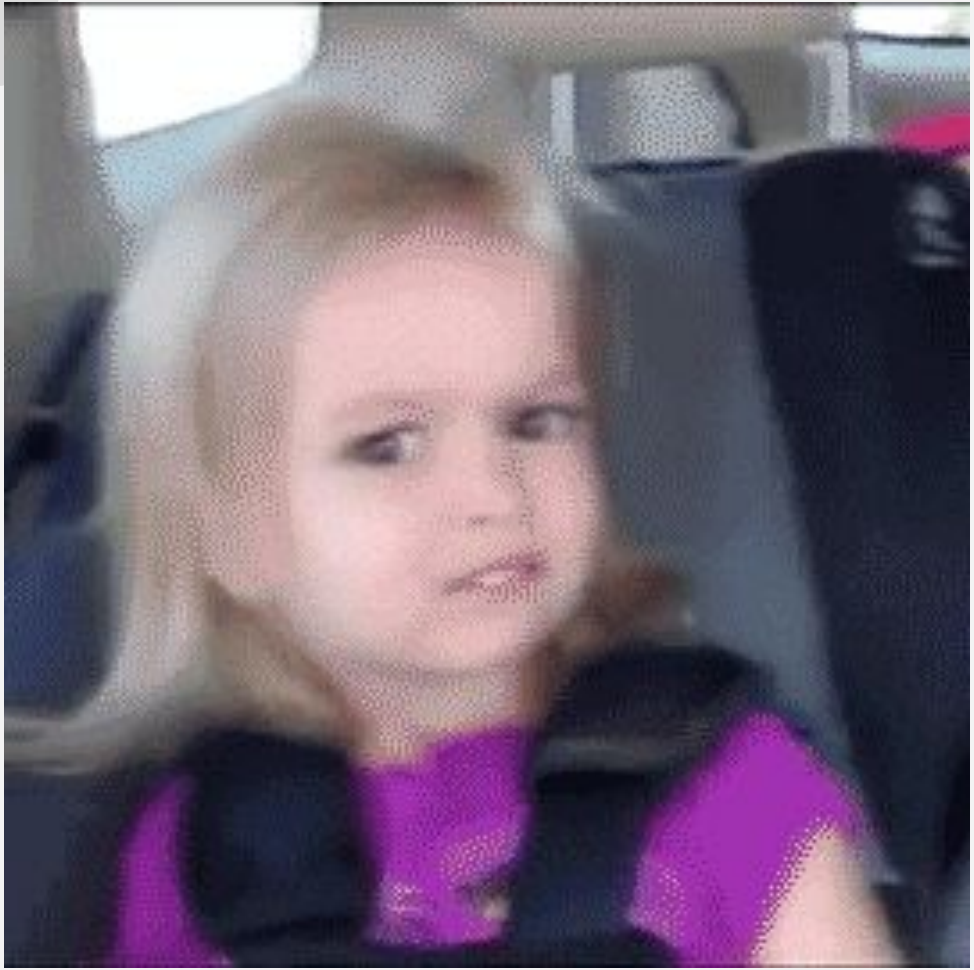
1. ***Class Decorators:*** Mark a class as a Controller, Service, or Module.

2. ***Method Decorators:*** Define HTTP methods (Get, Post) for functions.

3. ***Parameter Decorators:*** Extract data from the request (Body, Param, Query).

Why Decorators Matter

- Route definitions live with handlers
- Type-safe parameters
- Self-documenting
- Framework can analyze structure



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Separation of Concerns

The Three Layers

Separation of Concerns is a software design principle that says:

"Each part of your application should have a single, well-defined responsibility."

- Instead of mixing everything together, you split your code into layers, so each layer only does one thing.
- This makes your code: Clean, Maintainable, Testable, and Scalable .

The Three Layers

1. ***Controllers:***

   HTTP concerns only. Handle requests and send responses.

2. ***Services/Providers:***

   Business logic. Where the actual work happens.

3. ***Data Access:***

   Database operations.

The Three Layers

Controllers: HTTP Layer

Responsibility: Handle HTTP requests and responses ONLY.

Should NOT: Contain business logic, make database queries, or perform calculations.

Services: Business Logic Layer

Responsibility: Implement business rules and logic.

The @Injectable() Decorator: Marks a class as managed by the NestJS container.

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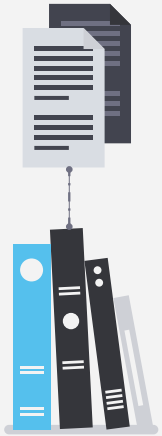

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Module System Architecture

What Are Modules?

Definition:

Organize related components (controllers, services) into cohesive blocks. Every NestJS application has at least one module (the root module).

Module Relationships

- Feature Modules: Encapsulate specific features (e.g., UserModule).
- Shared Modules: Share services (like DatabaseService) across the application.
- Global Modules: Available everywhere without importing (e.g., LoggerModule).

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Middleware

What is Middleware?

Definition:

Functions that execute before the route handler. They sit between the request and the controller.

Common Uses:

- Logging
- Parsing request bodies
- Authentication (basic)

Types in NestJS:

- Functional Middleware: Simple functions.
- Class-based Middleware: Classes that can inject dependencies.

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Guards

What Are Guards?

Definition:

Determine if a request should proceed based on conditions (auth, roles, etc.).

Key Difference from Middleware:

- Guards run after middleware
- Have access to execution context (knows what handler will be called)
- Specifically for authorization decisions

What Are Guards?

Common Guards:

- AuthGuard: Checks if a user is logged in.
- RolesGuard: Checks if a user has the required permission.

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pips

What are Pips?

Its job is to process incoming data before it reaches the controller.

Key points:

- validate, transform, or sanitize request data.
- Keeps controllers clean, because all input processing is handled here.

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Interceptors

What are Interceptors?

An Interceptor is a class that wraps around a controller method. Can run before and after the controller executes.

Key points:

Can log requests, measure execution time, cache responses, or transform output. Can inject other services/providers using DI.

Execution Order:

Request → Middleware → Guards → Interceptors (Before) → Pipes → Controller → Interceptors (After) → Service → response

When to Use What

Use Case	Tool
Logging requests	Middleware
Parsing body	Middleware
Authentication	Guard
Authorization (roles)	Guard
Validation	Pipe
Transform response	Interceptor

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Big Announcement!

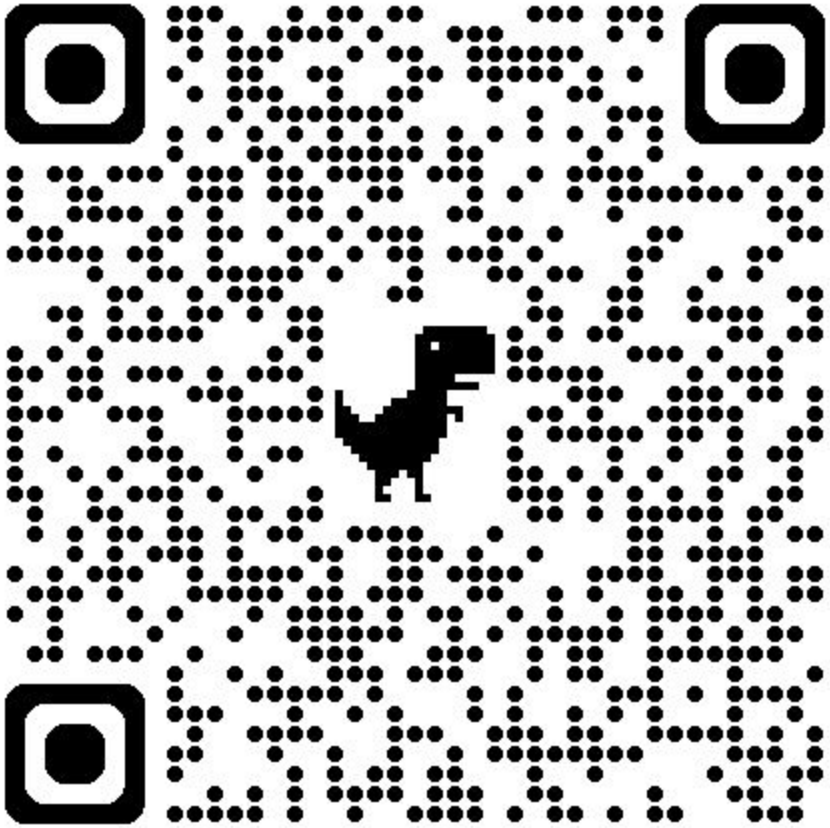
Who is best of the Week?

Game Time!

Mohsens Edition!

من سيربح المليون

Game!



1_ استعين بصديق

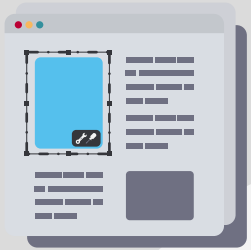
2_ حذف اجابتين

3_ تصويت الجمهور

theory mode: completed!

Time to roll up our sleeves and dive into the fun part

the technical part begins



Thanks!